

B. TECH
(SEM VII) THEORY EXAMINATION 2022-23
BIOSEPARATION & DOWN STREAM PROCESSING

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 10 = 20**

- (a) Discuss centrifugal force. Explain ultracentrifugation.
- (b) What are the two main processes involved in downstream processing?
- (c) Define isopycnic sedimentation.
- (d) Define Reynold's number
- (e) Differentiate between micro & ultra filtration.
- (f) Differentiate between flocculation and floatation. Give example.
- (g) Name different types of rotors. Explain any one in brief.
- (h) Define stabilization.
- (i) Discuss the factors affecting the characteristics of filters.
- (j) Name the material which can be used as Ion exchangers?

SECTION B

2. Attempt any three of the following:**10 x 3 = 30**

- (a) What is the need for Bioseparation? Discuss the Steps Associated with Downstream Processing.
- (b) Which biomolecules support all properties of life? How does the structure of biomolecules determine their properties and functions?
- (c) Write short note on:
 - i) Separation through ion- exchange chromatography
 - ii) Hybrid separation technologies
- (d) What factors affect centrifugation? Enlist the type of centrifugation used in biotechnology industry.
- (e) Differentiate between the working principles of Surface filter and depth filters. Also discuss their application.

SECTION C

3. Attempt any one part of the following:**10 x 1 = 10**

- (a) Discuss the principle of supercritical extraction of fluids. Add a note on super critical temperature.
- (b) What is formulation? How are final products formulated after lyophilisation?

4. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Discuss the principle of electrophoresis technique with reference to gel electrophoresis and isoelectric focusing.
- (b) Discuss the factors which affect isolation procedure by precipitation method. Explain the precipitation process using salts and organic solvents.

5. Attempt any *one* part of the following: 10 x 1 = 10

- (a) What do you understand by scale up? Discuss design of batch crystallizers.
- (b) Explain in detail ultra filtration, and reverse osmosis.

6. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Describe the design and working principle of freeze dryer for product polishing.
- (b) Discuss the principles of aqueous two-phase partitioning of proteins. Add a note on affinity partitioning.

7. Attempt any *one* part of the following: 10 x 1 = 10

- (a) Discuss the commercial importance of monoclonal antibodies.
- (b) Explain how GC-MS is used in the purification of bio-products.